

[ARCHIVED CATALOGUE]

Computer Engineering (MS)

A minimum grade point average of 3.0 must be earned on all course work applied toward the master's degree in electrical engineering. This average must also be achieved on all 400-level and above course work attempted at USC beyond the bachelor's degree and through an accumulation of no more than 12 units beyond the minimum needed for the specific degree program. Transfer units count as credit (CR) toward the master's degree and are not computed in the grade point average.

In addition to the general requirements of the Viterbi School of Engineering, the Master of Science in Computer Engineering is also subject to the following requirements: (1) a total of at least 28 units is required; (2) every non-EE course for graduate credit requires prior written adviser approval recorded each semester on a special request form in the student's department file; (3) no more than three courses (maximum 12 units) may be counted at the 400 level — at least 19 adviser-approved units must be taken at the 500 or 600 level; (4) at least 20 units must be taken in Computer Engineering; (5) a maximum of two [EE 599](#) Special Topics courses (8 units) may be counted toward the MSCENG; (6) at most, 4 units of [EE 590](#) (Directed Research) and 1 unit of seminar, [EE 598](#) (or other seminar with adviser approval), may be counted toward the MSCENG; (7) units to be transferred (maximum 4 with adviser approval) must have been taken prior to taking classes at USC — interruption of residency is not allowed.

In addition to the above, the Master of Science in Computer Engineering is earned by completing an integrated program of at least 28 units of approved course work emphasizing three key areas - Architecture, VLSI and Networks. Students must take one course from at least two areas, outlined below.

For Admission Requirements, refer to [Viterbi Graduate Degrees and Requirements](#).

Computer Engineering course work requirements

Take a minimum of 20 units of Computer Engineering courses from the three lists below, including at least one course from two of the three areas in Architecture, Networks and VLSI/CAD areas. Electives from the Foundations area will also count toward the 20-unit Computer Engineering requirement.

Computer Architecture

Take at least one course from two areas.

- [EE 451 Parallel and Distributed Computation](#) Units: 4
- [EE 454L Introduction to System-on-Chip](#) Units: 4
- [EE 457 Computer Systems Organization](#) Units: 4
- [EE 533 Network Processor Design and Programming](#) Units: 4
- [EE 542 Internet and Cloud Computing](#) Units: 4
- [EE 557 Computer Systems Architecture](#) Units: 4
- [EE 560L Digital System Design](#) Units: 4
- [EE 568 Parallel Programming](#) Units: 4
- [EE 638 Application of Machine Learning in the Real World](#) Units: 4
- [EE 653 Advanced Topics in Microarchitecture](#) Units: 3
- [EE 657 Parallel and Distributed Computing](#) Units: 3
- [EE 659 Advanced Topics in Interconnection Network Design and Analysis](#) Units: 4
- [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2

Networks

Take at least one course from two areas.

- [EE 450 Introduction to Computer Networks](#) Units: 4
- [EE 467 Introduction to Communication Systems](#) Units: 4
- [EE 533 Network Processor Design and Programming](#) Units: 4
- [EE 550 Data Networks: Design and Analysis](#) Units: 4
- [EE 555 Broadband Network Architectures](#) Units: 3
- [EE 597 Wireless Networks](#) Units: 4
- [EE 650 Advanced Topics in Computer Networks](#) Units: 3

VLSI/CAD

Take at least one course from two areas.

- [EE 477L MOS VLSI Circuit Design](#) Units: 4
- [EE 536a Analog Integrated Circuits](#) Units: 4
- [EE 536b Analog Integrated Circuits](#) Units: 4
- [EE 537 Modern Solid-State Devices](#) Units: 4
- [EE 552 Asynchronous VLSI Design](#) Units: 4
- [EE 554 Cyber-Physical Systems: A Computing Perspective](#) Units: 4
- [EE 576 Integrated Memory Devices and Technologies](#) Units: 4
- [EE 577a VLSI System Design](#) Units: 4
- [EE 577b VLSI System Design](#) Units: 4
- [EE 578 Computational Electromagnetics for Engineers](#) Units: 4
- [EE 580 System Verification](#) Units: 4
- [EE 581 Mathematical Foundations for System Design: Modeling, Analysis, and Synthesis](#) Units: 4
- [EE 582 CMOS:Nano Neuromorphic Circuits](#) Units: 4
- [EE 658 Diagnosis and Design of Reliable Digital Systems](#) Units: 4
- [EE 677 Accelerated Computing using Field Programmable Gate Arrays](#) Units: 2
- [EE 680 Computer-Aided Design of Digital Systems](#) Units: 4

Foundations

Additional approved Computer Engineering electives that will count toward the 20 units required in the computer engineering area.

- [CSCI 402 Operating Systems](#) Units: 4
 - [CSCI 570 Analysis of Algorithms](#) Units: 4
 - [EE 503 Probability for Electrical and Computer Engineers](#) Units: 4
 - [EE 510 Linear Algebra for Engineering](#) Units: 4
 - [EE 541 A Computational Introduction to Deep Learning](#) Units: 2
 - [EE 559 Machine Learning I: Supervised Methods](#) Units: 4
 - [EE 588 Optimization for the Information and Data Sciences](#) Units: 4
 - [EE 595 Software Design and Optimization](#) Units: 4
 - [EE 660 Machine Learning II: Mathematical Foundations and Methods](#) Units: 4
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